

Enhancing LLM’s capacity to design and solve games in the style of NYT’s “Connection” game

Anonymous NAACL-SRW 2025 submission

Abstract

Word games like the New York Times’ Connections puzzle, which challenge players to group words by shared themes, offer a unique opportunity to test linguistic understanding and abstract reasoning in large language models (LLMs). Connections was chosen for its emphasis on semantic relationships and nuanced reasoning, making it a valuable benchmark for evaluating LLM performance. This paper explores generating Connections puzzles and improving LLM solve rates using the Tree of Thoughts (ToT) framework. ChatGPT-4-Turbo achieved a 23.46% solve rate on high-difficulty puzzles, compared to GPT-3.5 Turbo’s 1.73%, highlighting gaps in semantic reasoning.

Beyond solving puzzles, this work contributes to procedural content generation, extending replayability and design principles to other word games. It also sheds light on LLM strengths and limitations, offering actionable insights into prompt engineering, puzzle complexity, and future directions for advancing AI-driven abstract reasoning.

1 Introduction

Word games, whose mechanics involve manipulating, parsing, or inventing words, have a long history, dating back to at least the Roman era (Raphael, 2020). The recent surge in the popularity of online word games, including those published by the New York Times, underscores their enduring appeal. Despite some research on solving human authored word games, generating novel word games has received limited attention from AI researchers (Merino et al., 2023). In this paper we focus on “Connections”, a daily word puzzle game published by the New York Times since June 12,

2023, challenges players to categorize a grid of sixteen words into four groups of four related words each, based on shared themes or clues (Todd et al., 2023). The game’s complexity varies, with simpler groups like types of fish (bass, flounder, salmon, trout) to more complex groups such as words that form phrases starting with "fire" (ant, drill, island, opal). Solving a Connections puzzle requires understanding both the literal meanings and contextual subtleties of the words, as well as recognizing non-semantic properties, like words that look the same when rotated (e.g., MOW and NOON). This makes the puzzle a test of both linguistic understanding and abstract reasoning (Todd et al., 2023).

Large language models (LLMs) have shown significant success in capturing linguistic patterns from extensive corpora of human-produced text, performing well across various domains (Brown et al., 2020; Radford et al., 2019; Devlin et al., 2018; Vaswani et al., 2019). Recent research has begun to explore the capacity of LLMs for multi-step abstract reasoning (Borgeaud et al., 2022; Brown et al., 2020). We propose methods for generating complex “Connections” puzzles and improving LLM solve rates using ToT. Initial findings suggest that while these methods can solve the puzzles a substantial proportion of the time, they achieve low accuracy, especially at the more complex levels e.g Chat-GPT Turbo (CoT) with a success rate of 23.46% (Merino et al., 2024). This performance gap offers a solid ground for studying the successes and failures of LLM’s in encoding and retrieving semantic information as well as examining the impact of different prompts on LLM performance and exploring more challenging puzzle variants.

The impact of our work is multifaceted. First, we provide a new source of puzzles for a game with a limited number of published puzzles. Additionally, our prompting approach offers a framework

applicable to other word games, paving the way for procedural content generation. By studying puzzle generation methods, we gain insights into the design of Connections puzzles and observe the effects of various strategies on user experience. Finally, our experiments reveal the capabilities and limitations of underlying LLMs, highlighting valuable directions for future research.

2 Methodology

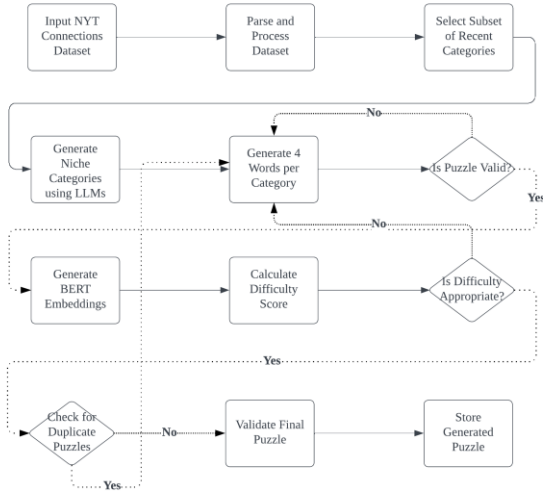


Figure 1: Flowchart of the puzzle generation

2.1 Identifying puzzle constraints

Although there are no “set” given rules by the New York Times for the making of these puzzles, from analysis of their games as well constraints set by (mention authors name) we can infer the following fundamental criteria, applicable to all of their games in the style of “Connections”:

1. The puzzle contains 16 unique words.
2. The puzzle has 4 unique groups of 4 words.
3. All words in a group share a connection, which must be unique to that set of words only in the puzzle.
4. Each word is used exactly once.
5. The puzzle has exactly one solution.
6. Categories are more specific than 5-LETTER WORDS, NAMES, or VERBS.
7. Puzzle needs to be Valid
8. The words in each category must be unique to the category connection (e.g the word GREEN will not appear in the category SHADES OF GREEN)

9. Each of the word’s spellings matter (i.e. BE will not appear in the category INSECT, but BEE might)
10. The Connection, between each set of 4 words must be of increasing difficulty

We introduce advanced strategies to increase puzzle difficulty, including ambiguous categories, complex word relations, and broader associations.

2.2 Added Level of Difficulty

To enhance the complexity of the puzzles, we introduce the following advanced strategies:

- (1) Ambiguity with Homophones and Homographs: Words with identical spelling or pronunciation but different meanings (e.g., "BARK" as a tree covering or a dog's noise) challenge solvers to rely on contextual clues for interpretation.
- (2) Obscure or Less Known Connections: Incorporating categories that require lateral thinking or niche knowledge, such as ancient mythology or scientific terminology, increases difficulty and encourages solvers to broaden their understanding.
- (3) Layering of Multiple Semantic Fields: Puzzles may include overlapping categories from diverse fields (e.g., "MUSIC" and "CHEMISTRY"), requiring solvers to distinguish between subtle, non-intuitive relationships.
- (4) Dynamic Difficulty Adjustment: Based on solver performance, puzzles are tailored to gradually introduce more complex categories, ambiguous connections, and nuanced word relationships.
- (5) Contextual Wordplay: Words with figurative meanings or puns (e.g., "FALL" as a season or "THINGS THAT FALL") prompt solvers to think laterally and consider multiple interpretations.

By applying these strategies, our puzzles aim to provide a more intellectually stimulating experience, equipping solvers with the skills to tackle increasingly complex challenges in games like the New York Times’ “Connections.”

2.3 Measuring Difficulty

We evaluated puzzle difficulty using a similarity-based scoring approach with word embeddings.

The difficulty of a word group was defined as the average semantic similarity between all word pairs within the group. Higher similarity scores indicate closely related words, making the group easier to identify, while lower scores reflect greater semantic diversity, increasing difficulty.

Word embeddings were generated using a pre-trained BERT model in a zero-shot setting, with each word represented as a high-dimensional vector. Pairwise cosine similarity scores were calculated and averaged for each group to produce a single difficulty score. The overall puzzle difficulty was determined by averaging the scores of all groups within the puzzle.

For example, consider the following results from Puzzle 10:

FRUIT (Difficulty: 0.493):
Date, Kiwi, Lemon, Orange
COUNTRIES (Difficulty: 0.484):
Chad, Georgia, Jordan, Togo
BIRDS (Difficulty: 0.441):
Crane, Jay, Swallow, Turkey
ZODIAC SYMBOLS (Difficulty: 0.423):
Fish, Goat, Scales, Twins

In this example, the "FRUIT" group has the highest difficulty score (0.493), indicating closer semantic relationships and easier identification. Conversely, "ZODIAC SYMBOLS" has the lowest score (0.423), reflecting greater semantic diversity and increased difficulty. This method provides a quantitative way to compare puzzles and assess their relative complexity, enabling objective evaluation of puzzle design.

2.4 Puzzle Generation Pipeline

Dataset: The initial dataset was sourced from a JSON file (connections.json) containing existing categories and words used in previous NYT's official 400 connections games. This dataset was parsed and processed to extract relevant categories and associated words.

To evaluate the model's performance, we created two separate datasets for training and testing. Using GPT-4o, we generated a total of 2,000 challenging training datasets and 400 challenging testing datasets. Each puzzle dataset consists of 4-word groups, with each group containing 4 words. This results in a total of 9,600 words across all datasets.

These datasets were specifically designed to test the model's ability to generate coherent and challenging puzzles, with varying levels of difficulty.

Category Limitation: To maintain computational efficiency and avoid hitting the token limit, only a subset of the most recent or relevant categories was selected from the existing dataset. This subset was limited to the last 50 categories.

Niche Category Generation: The OpenAI GPT-4o model was utilized to generate new, challenging categories. The prompt was carefully crafted to ensure the generated categories were niche, expansive, and more specific than basic categories such as "5-LETTER WORDS," "NAMES," or "VERBS." This process involved feeding the existing categories into the model and asking for four new categories that fit the criteria.

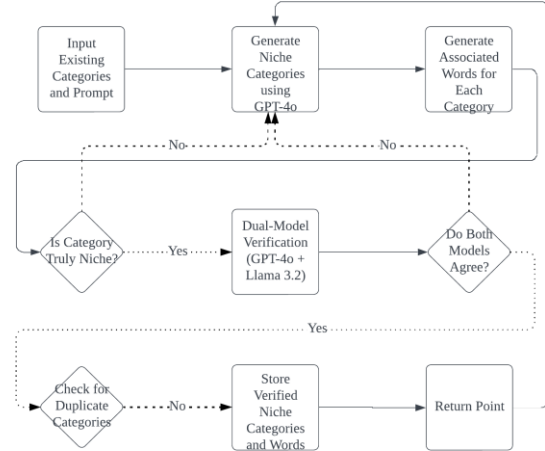


Figure 2: Flowchart of the Niche Category Generation

To determine whether a category is truly "niche," we applied a dual-model verification process. After GPT-4o generated the categories and associated words, we used both GPT-4o and Llama 3.2 8B models to independently evaluate whether the category could be considered niche (binary decision: Yes/No). A category was defined as niche only if both models agreed on its classification. This approach ensured consistency and reduced potential bias from a single model's perspective.

Word Generation for Categories: For each new category, four words were generated that fit the category perfectly. The prompt guided the model to ensure that the words were connected in a non-

obvious way, considering different uses, spellings, or meanings.

Validation of Puzzle Integrity: To ensure the validity of the generated puzzle, a validation function was implemented. This function checks that

- (1) All words are unique.
- (2) The puzzle has exactly four groups, each containing four words.
- (3) Additional validation criteria can be added as needed.

Embedding Generation: To evaluate how closely words align with their respective categories, we used a BERT model (bert-base-uncased) to generate word embeddings via the [CLS] token. BERT's bidirectional processing captures contextual relationships between words, enabling rich, context-aware embeddings. These embeddings effectively measure semantic similarity, ensuring words fit their categories while capturing subtle, non-obvious connections to create more challenging puzzles.

3 Results

3.1 Dataset Difficulty

To quantitatively evaluate the difficulty of word-group puzzles, we employed a SentenceTransformer model (all-mpnet-base-v2) to compute semantic embeddings for each word group. Difficulty was assessed based on intra-group semantic similarity, with lower average similarity scores indicating harder puzzles. This method allows for a scalable and objective evaluation of puzzle complexity across datasets.

The difficulty of each word group was calculated as follows:

- (1) **Embedding Calculation:** Each word in a group was encoded into a 768-dimensional vector using the SentenceTransformer model.
- (2) **Cosine Similarity Matrix:** Pairwise cosine similarity scores were computed between all word embeddings within the group, forming a similarity matrix.
- (3) **Average Similarity:** The mean of all similarity scores in the matrix was used as the difficulty score for the group. A lower

average similarity indicates that the words in the group are semantically diverse, making the puzzle harder to solve.

Formally, for a word group $G = \{w_1, w_2, \dots, w_n\}$, the difficulty score $D(G)$ is defined as:

$$D(G) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \cos_sim(\text{embed}(w_i), \text{embed}(w_j))$$

Where $\text{embed}(w)$ is the embedding of word w , and $\cos_sim$ is the cosine similarity between two embeddings.

For each puzzle, the average difficulty score was calculated by aggregating the scores of all word groups contained within the puzzle. The overall difficulty of a dataset was then computed as the mean difficulty score across all puzzles in the dataset.

The evaluation revealed notable differences in difficulty between human-crafted and AI-generated puzzles:

The real dataset scored an average difficulty of 85.52, indicating that human-crafted puzzles are generally easier to solve.

Among the AI-generated datasets, GPT-3.5 Turbo produced puzzles with similar ease (84.98), while GPT-4 variants, employing advanced reasoning frameworks such as Chain-of-Thought (CoT) and Tree-of-Thought (ToT), generated puzzles with significantly lower scores (77.08 and 75.02, respectively), reflecting increased complexity.

The following table summarizes the difficulty scores for each dataset:

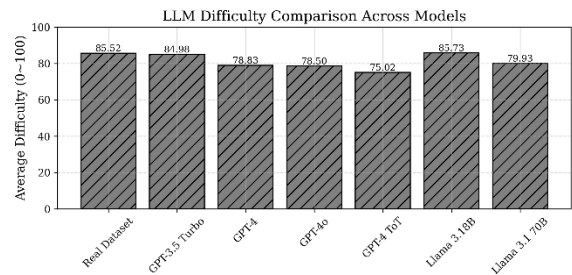


Figure 3: Solve Rates Across Models

The results indicate that human-crafted puzzles tend to group semantically similar words, resulting in higher average similarity scores and easier puzzles. In contrast, AI-generated puzzles, particularly those created using advanced reasoning frameworks, exhibit greater semantic

diversity within word groups, leading to lower similarity scores and increased difficulty. This suggests that LLMs, especially those employing structured reasoning methods, are capable of generating puzzles that challenge human solvers by introducing greater variability in semantic relationships.

3.2 Accuracy Metrics

The solve rates across models offer insight into puzzle tractability. Table 1 provides a detailed comparison of solve rates for both base and challenge variants across models.

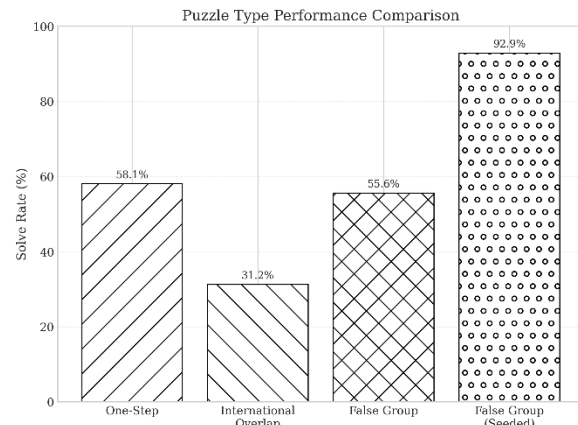


Figure 4: Solve Rates Across Models

Chain-of-Thought (CoT) prompting improved solve rates by an average of 9.73%, particularly for more complex puzzles. This underscores the effectiveness of structured reasoning in addressing the inherent complexity of AI-generated puzzles.

3.3 Performance by Puzzle Type

The study further categorizes puzzles into four types—"One-Step," "Intentional Overlap," "False Group (LLM)," and "False Group (Seeded)." Table 2 summarizes the solve rates for each type:

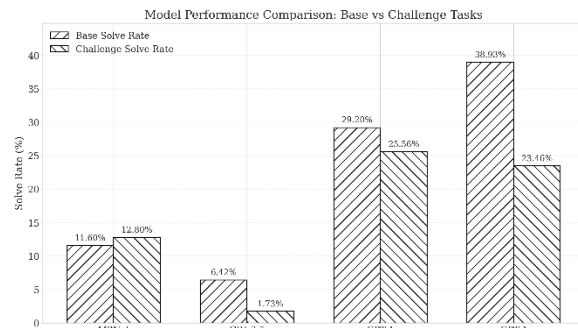


Figure 5: Solve Rates by Puzzle Type

4 Conclusion

This study addresses the challenge of evaluating and improving large language models (LLMs) in solving complex word puzzles, using the Connections game as a benchmark for linguistic understanding and abstract reasoning. By investigating LLMs' capabilities in both generating and solving puzzles, we highlight the potential of AI-driven reasoning frameworks like Tree-of-Thought (ToT) and Chain-of-Thought (CoT) in advancing procedural content generation (PCG).

Our findings demonstrate that AI-generated puzzles, leveraging ToT and CoT frameworks, achieve higher complexity than human-created ones, with difficulty scores as low as 67.61. CoT prompting improved solve rates to 38.93%, yet persistent challenges in nuanced semantic reasoning—particularly in advanced puzzle types like "Intentional Overlap"—underscore the need for more sophisticated reasoning frameworks.

Beyond Connections puzzles, this work contributes to the broader field of AI by advancing methods for abstract reasoning and semantic understanding. These findings have implications for educational tools, interactive game design, and other NLP applications requiring deliberate reasoning, such as multi-step question answering and document clustering. Future research should build on these insights by addressing current limitations, refining reasoning frameworks, and expanding datasets to further enhance AI's problem-solving capabilities across diverse linguistic and conceptual domains.

5 Limitations

While our study demonstrates promising results in generating and solving Connections puzzles, several limitations should be acknowledged. First, the solve rates of LLMs, particularly for puzzles with intentional overlaps and ambiguous categories, remain low, emphasizing the need for more advanced reasoning frameworks. Second, the dataset used in this study, while extensive, is limited to puzzles inspired by the New York Times' Connections game, which may restrict the generalizability of our findings to other word games or domains.

416 Additionally, our evaluation relies on semantic 464 *24th International Conference on Machine Learning*,
417 similarity metrics derived from pre-trained BERT 465 pages 33–40.
418 embeddings. While effective, these metrics may 466 Gusfield, D. 1997. *Algorithms on Strings, Trees, and*
419 not fully align with human intuition, particularly 467 *Sequences*. Cambridge University Press, Cambridge,
420 for puzzles involving subtle contextual nuances or 468 UK
421 wordplay. Lastly, while the Tree of Thoughts 469
422 (ToT) framework proved effective in this context,
423 its applicability to other complex reasoning tasks 470
424 remains to be validated.

425 Future work should address these limitations by
426 exploring alternative evaluation metrics that
427 better reflect human intuition, expanding the
428 dataset to include a more diverse range of puzzles,
429 and refining the ToT framework to enhance its
430 robustness and generalizability for solving
431 complex linguistic and conceptual tasks.

432 6 References

- 433 Merino, T., Earle, S., Sudhakaran, R., Sudhakaran, S., &
434 Togelius, J. 2024. *Making New Connections: LLMs as*
435 *Puzzle Generators for The New York Times' Connections*
436 *Word Game*. arXiv preprint, arXiv:2407.11240v1.
- 437 Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. 2019.
438 *BERT: Pre-training of Deep Bidirectional Transformers*
439 *for Language Understanding*. arXiv preprint,
440 arXiv:1810.04805.
- 441 Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao,
442 Y., & Narasimhan, K. 2023. *Tree of Thoughts:*
443 *Deliberate Problem Solving with Large Language*
444 *Models*. arXiv preprint, arXiv:2305.10601.
- 445 Gallotta, S., Cerny, D., Togelius, J., & Zambetta, F. 2024.
446 *MarioGPT: Fine-Tuning GPT-2 for Procedural Content*
447 *Generation in Super Mario Bros*. *Proceedings of the*
448 *AAAI Conference on Artificial Intelligence and*
449 *Interactive Digital Entertainment*.
- 450 Jaramillo, D., Tang, E., Bhattacharyya, R., Griffiths, T.
451 L., & Narasimhan, K. 2024. *Solving Codenames with*
452 *Language Models: A Comparative Study*. *Proceedings*
453 *of the Conference on Empirical Methods in Natural*
454 *Language Processing (EMNLP)*.
- 455 Rasooli, M. S., & Tetreault, J. R. 2015. Yara Parser: A
456 Fast and Accurate Dependency Parser. Computing
457 Research Repository, arXiv:1503.06733. Version 2.
- 458 Ando, R. K., & Zhang, T. 2005. *A Framework for*
459 *Learning Predictive Structures from Multiple Tasks and*
460 *Unlabeled Data*. *Journal of Machine Learning Research*,
461 6:1817–1853.
- 462 Andrew, G., & Gao, J. 2007. *Scalable Training of L1-*
463 *Regularized Log-Linear Models*. In *Proceedings of the*